

$$\begin{aligned}
S &= \lambda xyz.xz(yz) \\
K &= \lambda xy.x \\
SKK &= (\lambda xyz.(xz) (yz)) (\lambda xy.x) (\lambda xy.x) = \\
&= (\lambda yz.((\lambda xy.x)z) (yz)) (\lambda xy.x) = \\
&= \lambda z.((\lambda xy.x)z) ((\lambda xy.x)z) = \\
&= \lambda z.(\lambda y.z) ((\lambda xy.x)z) = \\
&= \lambda z.(\lambda y.z) (\lambda y.z) = \\
&= \lambda z.z
\end{aligned}$$